KG-EDAS: A Meta-Metric Framework for Evaluating Knowledge Graph Completion Models

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Problem Statement

- **Many Methods:** Contradictory rankings across metrics & datasets prevent top model selection
- **Many Metric:** No consensus exists on which metric (MRR? Hits@1? MR?) should be prioritized

Methodology

This study evaluates eight Multi-Criteria Decision-Making (MCDM) aggregators to reliably rank Knowledge Graph Completion (KGC) models across multiple metrics (MRR, Hits@1/3/10, MR).

- **Evaluation based on Distance from Average Solution (EDAS):** It is a cardinal approach that measures positive/negative deviations from average performance.

$$Avg_j = \frac{1}{n}\sum_{i=1}^{n} X_{ij} \quad j = 1 \dots m, \quad PDA_{ij} = \frac{\max(0, x_{ij} - Avg_j)}{Avg_j}, \quad NDA_{ij} = \frac{max(0, Avg_j - X_{ij})}{Avg_j} \quad \textit{(Beneficial)}$$

$$PDA_{ij} = \frac{\max(0, Avg_j - X_{ij})}{Avg_j}, \quad NDA_{ij} = \frac{max(0, X_{ij} - Avg_j)}{Avg_j} \quad \textit{(Non-Beneficial)}$$

$$WPDA_j = \sum_{j=1}^{m} w \,.\, PDA_{ij}, WNDA_j = \sum_{j=1}^{m} w \,.\, NDA_{ij}, \quad N(WPDA_i) = \frac{WPDA}{max_i\,(WPDA_i)}, N(WNDA_i) = \frac{WNDA}{max_i\,(WNDA_i)},$$

$$M_i = \frac{1}{2}[N(WPDA)_i + (1 - (WNDA_i))]$$

- **Z-Score (ZS):** It standardizes metrics using mean and standard deviation.

$$Z_{ij} = \frac{X_{ij} - \mu_j}{\sigma_j}, \qquad ZS_i = \frac{1}{m}\sum_{j=1}^{m} z_{ij}$$

- We additionally consider six other established MCDM techniques: **TOPSIS, VIKOR, WASPAS, MOORA, Borda Count, and Pareto Frontier** to ensure comprehensive coverage of aggregation philosophies (cardinal, ordinal, and set-based).

Multi-Dimensional Testing

This pairing enables a principled comparison across four key components:

- **Consistency:** Alignment with base metrics (τ, r, ρ correlation)
- **Stability:** Cross-dataset ranking variance
- **Robustness:** Sensitivity to metric removal (leave-one-out)
- **Generalizability:** Performance on unseen benchmarks.

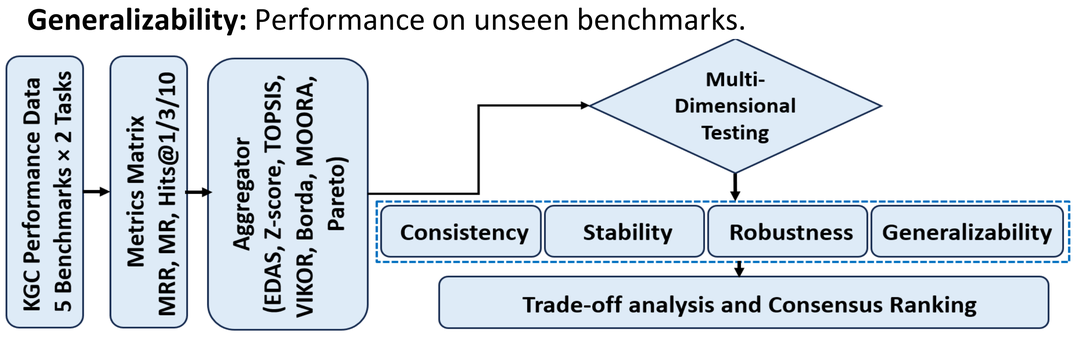

Fig 1: Methodology Diagram

Table 1: Link Prediction Results

	FB15k				WN18				FB15k-237				WN18RR				YAGO3-10			
Models	**MR**	**MRR**	**H@1**	**H@10**	**MR**	**MRR**	**H@1**	**H@10**	**MR**	**MRR**	**H@1**	**H@10**	**MR**	**MRR**	**H@1**	**H@10**	**MR**	**MRR**	**H@1**	**H@10**
RotatE	42	0.791	0.739	0.881	274	0.949	0.943	0.96	178	0.336	0.238	0.531	3318	0.475	0.426	0.573	1827	0.498	0.405	0.671
TuckER	39	0.788	0.729	0.889	510	0.951	0.946	0.958	162	0.352	0.259	0.536	6239	0.459	0.43	0.514	2417	0.544	0.466	0.681
RSN	70	0.773	0.706	0.886	471	0.95	0.946	0.959	251	0.346	0.256	0.526	5646	0.467	0.437	0.527	2582	0.527	0.446	0.673
ConvR	51	0.777	0.723	0.87	346	0.928	0.912	0.951	248	0.28	0.198	0.444	4210	0.395	0.346	0.483	1339	0.511	0.427	0.664

Results

Test 1: Consistency Analysis

- EDAS and TOPSIS show the highest overall alignment with standard metrics (*~0.6–0.8* with H1, H10, MR, MRR).
- VIKOR and WASPAS lowest correlated alignment.
- Rotate achieved the top average ranking performance (average rank *1.50*).

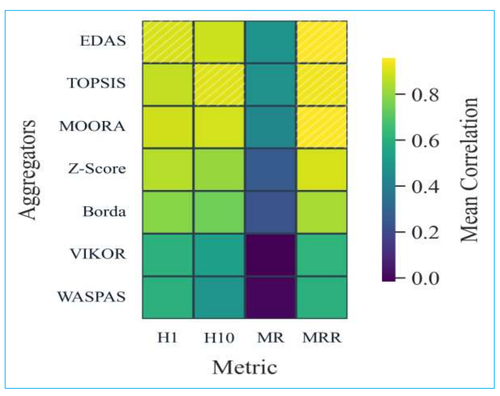

Fig 2: Variance Distributions Across Aggregators

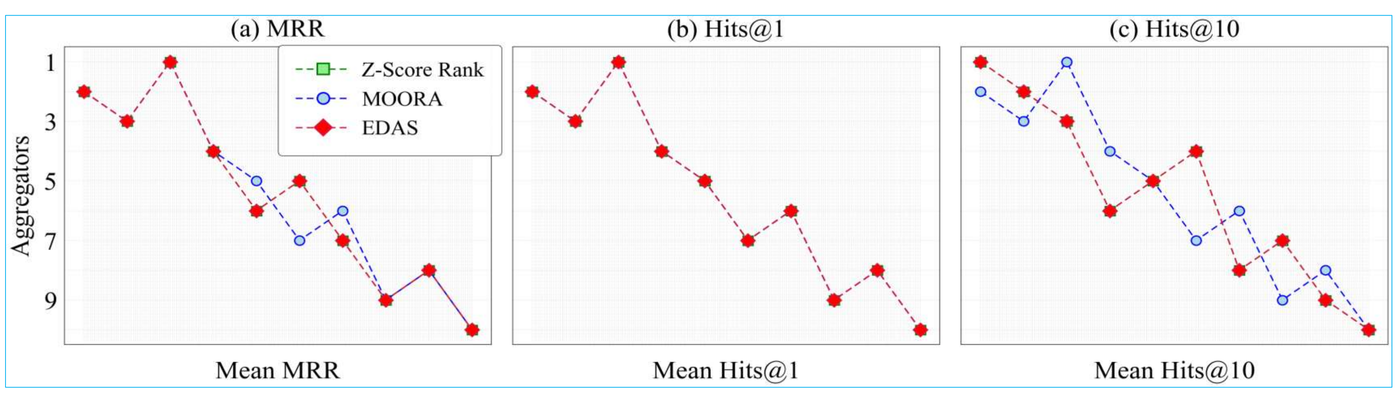

Fig 3: Correlation Among some Aggregators and Metrics

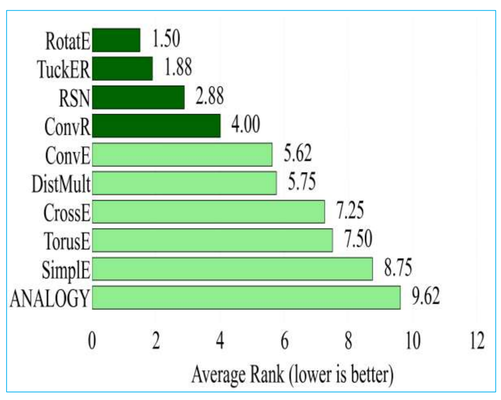

Fig 2: Average Ranking performance

Test 2: Stability Analysis

- VIKOR Q and Borda are most stable ranking (mean variance *2.88 and 2.912*).
- TOPSIS exhibits the highest instability (mean variance *4.016*).

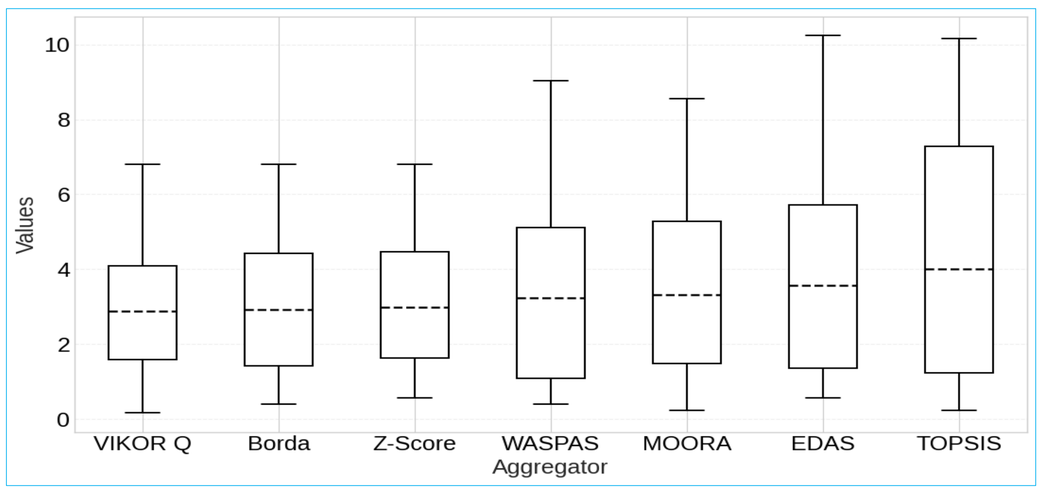

Fig 5: Distributions Variance values

Test 3: Robustness Analysis

- MR is Higher disruptive metric in the robustness analysis.
- Baseline and ablated rankings correlation *(> 0.90*, MRR, H@1/1-removed)

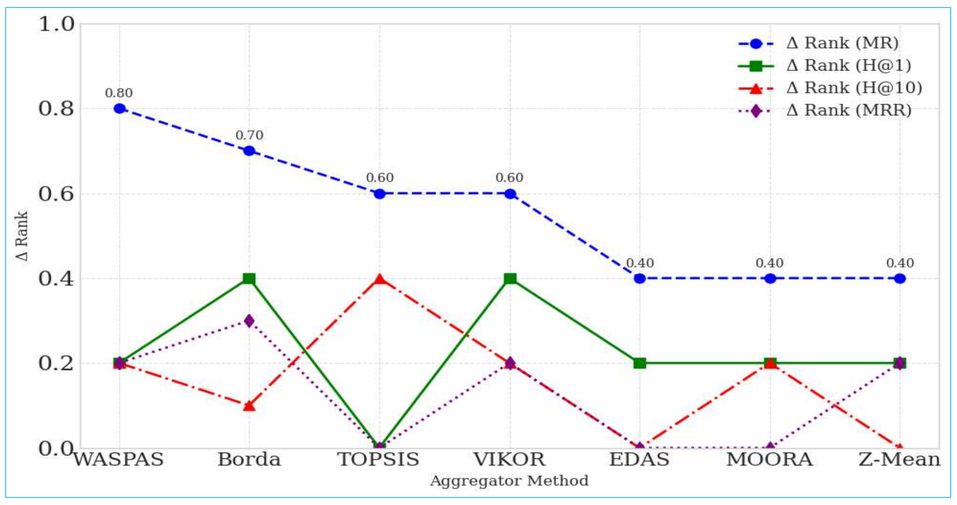

Fig 6: Robustness: MR most Disruptive Removal

Test 4: Generalization Analysis

- TOPSIS and WASPAS top-1 accuracy (*0.35*) higher than others. While Borda and Z-score top-3 accuracy (*0.95*).

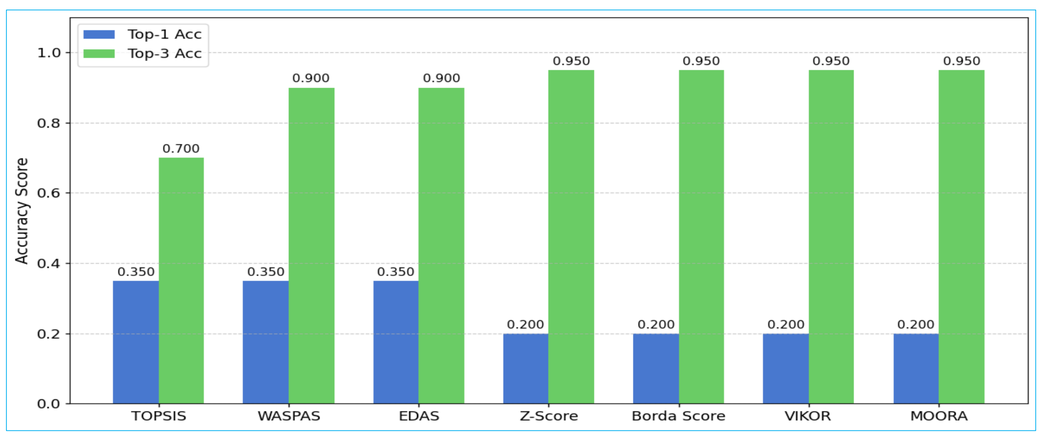

Fig 7: Generalization Top-1 and 3 Accuracy

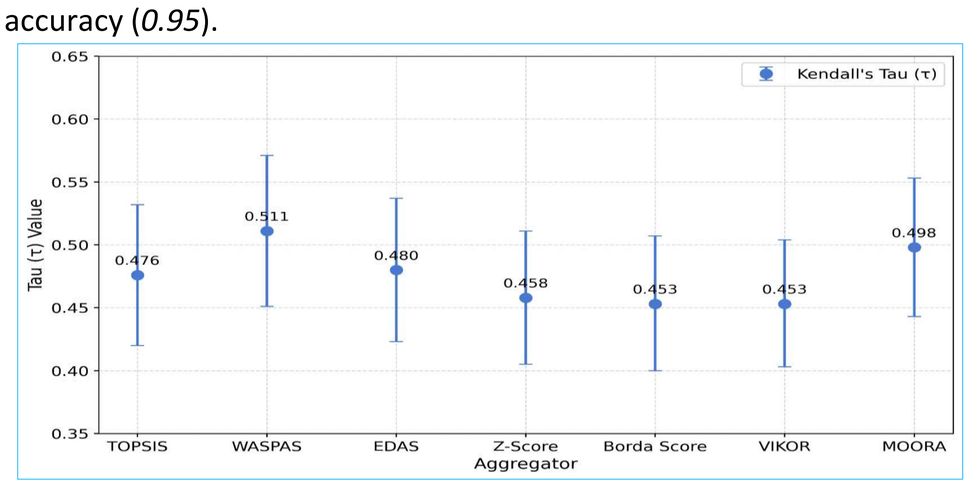

Fig 8: (τ) with 95% CI across MCDM Aggregators

Conclusion

- **(1) Trade-offs exist.** No single aggregator is best for all evaluation goals. **(2) Use EDAS or Z-Score** for robust, consistent overall rankings.
- **(3) Borda Count** is best for stable rankings across datasets., **(4) Consensus bests:** RotatE/TuckER (relation), MuCo-KGC/KICGPT (tail).
- **(5) Model stability varies greatly** (e.g., RSN stable, DistMult volatile).